Attribute-Cooperated Convolutional Neural Network for Remote Sensing Image Classification

Yuanlin Zhang, Xiangtao Zheng, *Member, IEEE*, Yuan Yuan, *Senior Member, IEEE*, and Xiaoqiang Lu, *Senior Member, IEEE*

***Abstract*— Remote sensing image (RSI) classification is one of the most important fields in RSI processing. It is well known that RSIs are very complicated due to its various kinds of contents. Therefore, it is very difficult to distinguish different scene categories with similar visual contents, like desert and bare land. To address hard negative categories, an attribute-cooperated convolutional neural network (ACCNN) is proposed to exploit attributes as additional guiding information. First, the classification branch extracts convolutional neural network feature, which is then utilized to recognize the RSI scene categories. Second, the attribute branch is proposed to make the network distinguish scene categories efficiently. The proposed attribute branch shares feature extraction layers with the classification branch and makes the classification branch aware of extra attribute information. Finally, the relationship branch constraints the relationship between the classification branch and the attribute branch. To exploit the attribute information, three attribute-classification data sets are generated (AC-AID, AC-UCM, and AC-Sydney). Experimental results show that the proposed method is competitive to state-of-the-art methods. The data sets are available at https://github.com/CrazyStoneonRoad/Attribute-Cooperated-Classification-Data sets.**

***Index Terms*— Attribute learning, convolutional neural networks (CNNs), relationship learning, remote sensing image (RSI) classification.**

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Yuanlin Zhang is with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China.

Xiangtao Zheng and Xiaoqiang Lu are with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China (e-mail: xiangtaoz@gmail.com).

Yuan Yuan is with the Center for Optical Imagery Analysis and Learning, School of Computer Science, Northwestern Polytechnical University, Xi'an 710072, China.

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I. Introduction

GIVEN a remote sensing scene, remote sensing image (RSI) classification tries to figure out the scene category. As a classical image processing task in remote sensing, RSI classification has wide applications in object detection [1], [2] and change detection [3], [4]. In RSI classification, two steps are considered [5]: feature representation and category recognition. Feature representation extracts a feature vector to represent each image, which has received much attention.

Earlier RSI classification methods are based on manual feature representations, such as the scale-invariant feature transform (SIFT) [6], the local binary pattern (LBP) [7], the bag of visual words (BoVW) [8], and the latent Dirichlet allocation (LDA) [9]. Although these manual feature representations have made great breakthroughs, they cannot meet the requirements of practical applications [10]: cannot deal with the complicated background information.

Compared with these manual feature representations, deep neural networks (DNNs) can provide effective feature representations, without human interference. DNNs have achieved great progresses in different fields, such as semantic segmentation [11], object detection [1], video classification [12], and scene classification [13]. Recently, many DNN-based RSI classification methods have been proposed. According to whether using multiple features, these methods can be divided as *single feature methods* and *multiple feature methods*. *Single feature methods* exploit multilayer features to improve the feature representation with a single model. Hu *et al.* [14] integrated fully connected features and dense convolutional features at multiple scales to strengthen the representations. Li *et al.* [15] integrated multilayer features of the convolutional neural network (CNN) and this method showed competitive performance with the state-of-the-art methods. Du *et al.* [3] for the first time deeply fused deep networks and slow feature analysis to construct state-of-the-art time series RSI fine classification method, providing very competitive and promising results. *Multiple feature methods* improve classification performance by fusing various kinds of features. To deal with the large intraclass and lower interclass variations, Alshehhi *et al.* [16] proposed a CNN framework accompanied by low-level features of roads and buildings. Anwer *et al.* [17] proposed the TEX-Nets by combining LBP with DNNs. These DNN-based RSI classification methods deal with complicated imagery contents through feature extraction and feature fusion. Recent studies have demonstrated that multiple feature integrating

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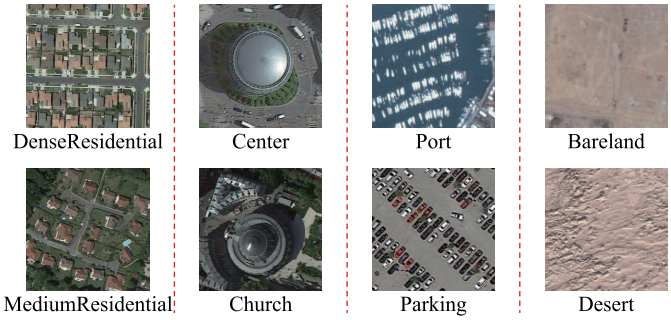


Fig. 1. Illustration of the visually fine-grained categories for RSI classification. Each column lists two categories of visually similar scenes.

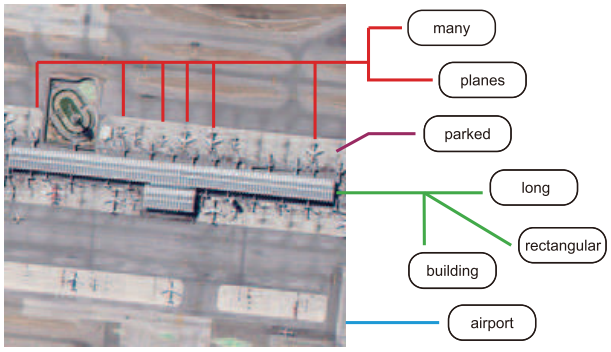


Fig. 2. Visualization of attributes of an image.

have great potential to enhance the discrimination capacity [17]. However, in all these methods, only the category information is used to generate scene representation, which is not sufficient to distinguish similar scene categories well. In RSI classification, most scene categories are similar in visual contents. These similar scene categories are defined as visually fine-grained categories. As shown in Fig. 1, it is quite common to see pairs of visually fine-grained categories. For example, *dense residential* is similar to *medium residential*; *center* is similar to *church*; *port* is similar to *parking*; and *bare land* is similar to *desert*.

To address similar scene categories, multitask learning (MTL) is considered. MTL uses knowledge contained in another or several other related tasks to improve generalization [18]. In this article, an attribute-cooperated convolutional neural network (ACCNN) is proposed to exploit attributes as additional guiding information. Fig. 2 presents the attributes of RSIs. It can be seen that a RSI can correspond to several attribute labels at the same time. Attributes can provide abundant details to distinguish the visually fine-grained categories [19]. Currently, there is not an RSI data set labeled with both scene categories and attributes. To avoid heavy labeling work, we figure out a way to generate new data sets based on existing data sets. It is found that there are several existing RSI caption data sets which partly correspond to some scene classification data sets. Words from these caption data sets are filtered, ordered, and selected to form the attribute labels of new attribute-cooperated scene classification data sets.

The proposed method consists of three aspects: the classification branch, the attribute branch, and the relationship branch. First, the classification branch predicts RSI scene categories and is the main branch of this method. This branch extracts DNN-based feature representation which is then utilized to recognize the RSI scene categories. Second, the attribute branch is proposed to make the classification branch aware of extra attribute information, which may distinguish the visually fine-grained scenes more efficiently. Finally, the relationship branch constraints the relationship between the classification branch and the attribute branch. The relationship branch further enhances the information sharing between different tasks and makes the classification branch aware of more additional attribute information.

In summary, the main contributions of this article consist of the following.

1) A novel RSI classification framework ACCNN is built. In this framework, attribute learning branch is introduced to help distinguish visually fine-grained categories, since the attribute branch can share extra information to the classification branch during learning.
2) The relationship learning branch is proposed to further share information of the attribute branch with the classification branch. Thus, the model can be more discriminating.
3) A fast and effective pipeline to build attribute-classification data sets is proposed. Three attribute-classification data sets are built based on the existing classification data sets and their corresponding caption labels.

The rest of this article is organized as follows. In Section II, some related works about this article are summarized. In Section III, the proposed method is presented. Section IV reports the experimental results. Section V makes the final conclusion.

II. Related Works

This section introduces some related studies. As mentioned earlier, the proposed classification framework involves different aspects of deep learning: MTL and *RSI classification*. Related works about these two aspects are illustrated as follows.

A. MTL

MTL refers to methods that use knowledge contained in another or several other related tasks to improve generalization. According to whether to use manual features, the previous methods are divided into nonneural network (NN) models and NN models.

In *non-NN models*, many existing methods made a sparse hypothesis about model parameters under the condition of closely related tasks. Argyriou *et al.* [20] deemed the shared features among all models as a small set and imposed least absolute shrinkage and selection operator (LASSO) on MTL to emphasize on this. To be mentioned, LASSO was also known as L_1 norm. Later, different kinds of mixed-norm constraints were implemented: L_1/L_∞ norm [21] and L_1/L_2 norm [20], which was also named as group LASSO. After this, many variations of the group LASSO were proposed [22]–[24].

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In addition, learning task relationships is important under the condition of nonclosely related tasks. Constraints that enforce a tasks clustering were proposed [25], [26]. Kim and Xing [27] extended the group LASSO for the tasks in a tree structure.

In NN models, several works have been done to investigate the MTL. Long and Wang [28] implemented a matrix prior on fully connected layers. Lu *et al.* [29] proposed an approach that can dynamically widen itself greedily during training using the criterion of similar task sharing features. Misra *et al.* [30] used soft parameter sharing between two separate models. Ruder *et al.* [31] generalized the aforementioned deep learning-based models. From these works, it can be seen that relationships between tasks must be considered to avoid sharing high-level semantic features among the nonclosely related tasks.

B. RSI Classification

It can be seen from the previous efforts [14]–[17] that feature extraction methods are the most investigated way to improve RSI classification. According to the feature semantic level, RSI classification development can be sorted into three stages as follows.

Low-level

1) First, the Low-Level Feature Methods: features contain color, texture, intensity, hue, saturation, and so on. Swain and Ballard [32] took color to provide visual skills for robots. Manjunath and Ma [33] used Gabor wavelet features to analyze texture information in RSIs. Ojala *et al.* [7] proposed the LBP in the RSI classification field. Lowe [6] proposed the SIFT feature which was invariant to variations like rotation, scaling, and intensity variation. Risojević and Z. Babić [34] proposed a nearest neighbor classifier based on local structural texture descriptor as well as a similarity measure.

2) Second, the Midlevel Feature Methods: Midlevel features are actually the capsulation of different attributes of low-level features. BoVW model [8] first extracted SIFT features from the image to learn visual words vocabulary and then encoded local descriptors against the vocabulary. Recently, Wang *et al.* [35] proposed a low rank representation (LRR)-based method. The semantic relationship among the visual words in BoVW was considered in LDA [9]. Probabilistic latent semantic analysis (pLSA) [36] was introduced to integrate multifeatures. Apart from supervised methods, weakly supervised methods [37] and unsupervised methods [38] are also investigated.

3) Third, the High-Level Feature Methods: High-level features mostly denote the deep learning methods at present. Deep learning-based methods used in RSI classification consists of autoencoder (AE) [39], deep belief network (DBN) [40], CNN [41]–[43], and so on. In the RSI classification field, the existing deep learning methods can be divided as two sets: methods with single layer of CNNs [44]–[49], and methods with multiple layers of CNNs [50]–[55].

1) *Methods With Single Layer:* The first set of methods took the fully connected layer or the soft-max layer. Othman *et al.* [46] extracted features from the last fully connected layer of a pretrained CNN and used an AE to implement RSI classification. In the work of Fang *et al.* [47], after tuning the CaffeNet, its soft-max layer was directly used to complete the classification task. Cheng *et al.* [48] introduced AlexNet, VGG-Net, and GoogLeNet to RSI classification. After this, Scott *et al.* [44] introduced CaffeNet, GoogLeNet, and ResNet. Then, Nogueira *et al.* [45] introduced AlexNet, CaffeNet, and OverFeat. Weng *et al.* [49] combined the AlexNet fully connected layer and the extreme learning machine. Wang *et al.* [56] proposed a collective feature fusion to get more robust representation to obtain better alignment.
2) *Methods With Multiple Layers:* The second set methods took features from different layers to improve expression ability. Penatti *et al.* [50] improved the classification through fusing fully connected layers of OverFeat and CaffeNet in a manner of cascade. Based on this work, Marmanis *et al.* [51] added a specially designed network where a soft-max layer output class labels. Chaib *et al.* [52] adopted discriminant correlation analysis (DCA) to fuse two fully connected layers of VGGNet. Inspired by BoVW, Cheng *et al.* [53] proposed to fuse multiple layers through bags of CNN features. Liu *et al.* [54] first proposed to combine convolutional layers and fully connected layers of CaffeNet in a linear manner. Wang *et al.* [55] proposed a method to fuse the hierarchical features from convolutional layers and fully connected layers. Zhang *et al.* [57] proposed the HRCNN and verified different functions of convolutional layers and fully connected layers.

III. ACCNN

The proposed method aims at dealing with the problem of complicated and visually fine-grained contents in the RSI classification task. Specifically, we propose an ACCNN to make the classifiers cooperate with the attribute learning. In this section, we introduce the ACCNN from five aspects: the framework, the classification branch, the attribute branch, and the relationship branch and the weights sharing designation.

A. Framework

As a kind of inductive transfer learning, MTL is helpful for two points [18]. First, MTL between very similar tasks enables the original model to acquire more general abilities and improves the classification performance in the testing phase. Second, there is a potential relationship between the tasks seeming irrelevant. This relationship can actually provide the model some latent basic skills. However, appropriate weights sharing should be designed and appropriate tasks should be selected to acquire these benefits.

1) Overall Description: As shown in Fig. 3, the left part of this figure is the convolutional layers which are shared in a hard-parameter-sharing way (see Section III-E). In the right part, three streams in blue, pink, and purple correspond to the branches of classification, attribute learning and relationship learning, respectively.

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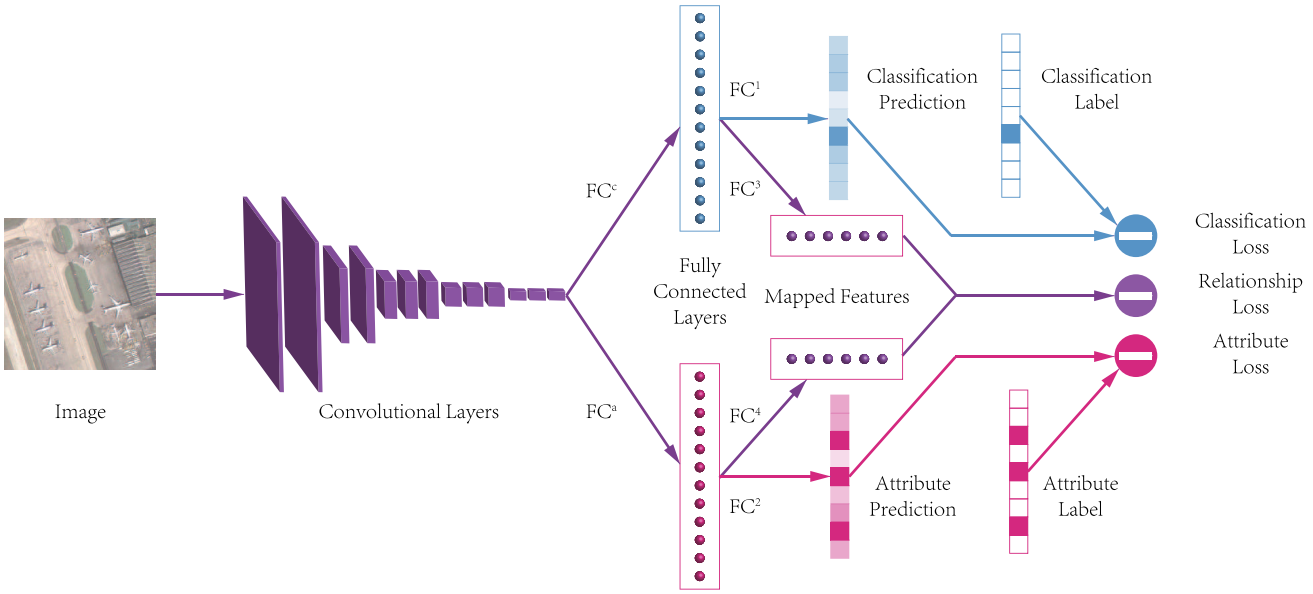


Fig. 3. Visualization of the proposed framework. (Left) Convolutional layers which are shared in a hard-parameter-sharing way. (Right) Three streams in blue, pink and purple correspond to the branches of classification, attribute, and relationship learning, respectively.

The proposed model starts with several groups of convolutional layers. As indicated in this figure, different groups of convolutional layers have various spatial scales. These convolutional layers correspond to the hard-parameter-sharing layers, built for both tasks of classification and attribute learning. The same features of these layers would be fed in both branches of the framework.

For the classification learning branch, a fully connected layer is utilized to acquire features for classification. The prediction result is a normalized vector where the elements sum to 1. Then, the classification prediction is compared with the classification label, a one-hot vector, to get the classification loss term (see Section III-B2).

For the attribute learning branch, another fully connected layer is utilized to acquire features for attribute prediction. The prediction result is a vector where each element values in the interval [0,1]. Then, the attribute prediction is compared with the attribute label, a binary vector with probably several true items, to get the attribute loss term (see Section III-C2).

For the relationship learning branch, two mapping layers, which are implemented with fully connected layers, are utilized to build a soft-parameter-sharing mechanism between high-level semantic features of the two branches. After this, an activation layer is used to normalize the outputs. Finally, these two layers are compared to get the relationship loss term (see Section III-D2).

2) Details: As shown in Fig. 4, the MTL is utilized to make the classification task cooperate with an auxiliary task. In our method, attribute learning is adopted as the auxiliary task of classification.

In the training phase, we take the parameters optimization algorithm, such as stochastic gradient descent (SGD), to train the model. The optimal objective function is formulated as

$$L = L_{\text{cls}} + \lambda_a L_{\text{atr}} + \lambda_r L_{\text{rlsn}} \tag{1}$$

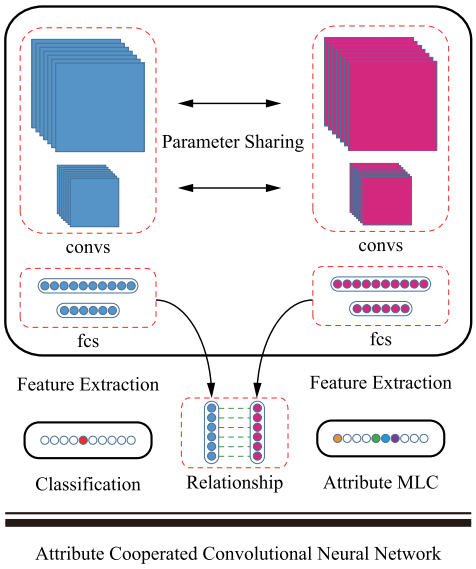


Fig. 4. Conceptual visualization of the proposed model. The ACCNN uses relationship learning to force the attribute multilayer classifiers to cooperate with the classification task.

where λ_a and λ_r represent the weights of different tasks. In this formula, L_{cls}, L_{atr}, and L_{rlsn} respectively represent the loss term of the classification branch, the attribute branch, and the relationship branch.

In the testing phase, our model can provide scene classification results together with the attribute information. Classification branch precision is expected to be improved compared with the classification task without any auxiliary task.

B. Classification Branch

1) Overall Description: The classification network consists of three parts: feature extraction, feature fusion (optional),

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and classification. The proposed model is based on a baseline without feature fusion. Feature extraction and classification are respectively described as follows.

As shown in the left column of Fig. 4, the feature extraction part indicates not only convolutional features but also fully connected features. Generally, deep convolutional layers have high-level semantic information and large receptive fields. However, shallow convolutional layers have the advantage of spatial resolution and have the potential to extract high-level semantic information. After the convolutional layers, several fully connected layers are attached to access global information.

The classification part usually means the last fully connected layer attached with a soft-max activation layer. The last fully connected layer converts the input features into a result vector, which is designed as a one-hot vector. In order to implement this, the result vector is activated by the soft-max layer to the form that sums to one. Finally, in the test phase, the top scored category of the result vector is taken as classification result.

2) Details: The classification network of our method first extracts convolutional features and fully connected features. Given the input image set $\mathrm{Im} = \{\mathbf{im}_m | m = 1, 2, \ldots, M\}$ as the first layer input $\mathbf{I}^0$, convolutional or fully connected layers are formulated as

$$\mathbf{I}^l = \mathrm{Activate}\left(\mathbf{W}^{l-1}\mathbf{I}^{l-1} + \mathbf{B}^{l-1}\right), \quad l = 1, 2, \ldots, L \tag{2}$$

where $\mathbf{W}^{l-1}$ and $\mathbf{B}^{l-1}$ denote weights and biases of the current layer, respectively. Activate() refers to the activation function which may vary in different CNNs. After this, the last layer features

$$\mathbf{I}^L = \left\{\mathbf{I}_m^L | m = 1, 2, \ldots, M\right\}, \quad \mathbf{I}_m^L = \left[z_1^m, z_2^m, \ldots, z_C^m\right]^{\mathrm{T}} \tag{3}$$

are fed into the classifiers and the output is obtained as

$$\mathbf{O} = \mathrm{SoftMax}\left(\mathbf{I}^L\right) = \{\mathbf{o}_m, | m = 1, 2, \ldots, M\} \tag{4}$$

where $\mathbf{o}_m = [p_1^m, p_2^m, \ldots, p_C^m]^{\mathrm{T}}$ is the predicted probability vector of the mth image, and the SoftMax function is formulated as

$$p_c^m = \mathrm{SoftMax}\left(\mathbf{I}_m^L\right) = \frac{e^{z_c^m}}{\sum_k e^{z_k^m}}. \tag{5}$$

During the training phase, the classification network is dedicated to minimize the loss

$$L_{\mathrm{cls}} = \mathrm{CrossEntropy}(\mathbf{O}) = \sum_m -\log\left(\frac{e^{z_{y_m}^m}}{\sum_c e^{z_c^m}}\right). \tag{6}$$

In this function, y_m is class index of the ground truth label of the mth image.

C. Attribute Branch

In order to improve the classification model, we need to make the scene classifiers cooperate with an appropriate task. Among all kinds of tasks (e.g., segmentation), attribute prediction is chosen for its unique superiority. Attribute prediction provides different kinds of high-level semantic tags for the same scene image. These attribute tags contain various aspects of detailed information, which provides a latent relationship between the scene and the attribute. In our model, the attribute tags are learned through multilabel classification (MLC) to make the shared weights general and robust.

1) Overall Description: The attribute network architecture is quite similar to the classification network and also consists of three parts: feature extraction, feature fusion (optional), and attribute prediction. Similarly, the attribute branch also does not have a feature fusion method.

As shown in the right column of Fig. 4, our attribute model has the same feature extraction framework with the classification model. The feature extraction part consists of both convolutional layers and fully connected layers. The convolutional layers of the attribute branch are directly shared with those of the classification branch. These layers are shared through a hard-parameter-sharing way (see Section III-E), which means the layers of different branches use totally the same parameters. The fully connected layers of these two branches are not directly shared. Actually, a relationship branch which would be illustrated later is proposed to share fully connected layers.

The attribute prediction is quite different from the classification branch. This prediction is implemented with multilabel classifiers (MLCs). Apparently, rather than being a one-hot vector, each predicted vector of MLC can have several positive elements. Correspondingly, the loss function of predictors is a binary cross entropy (BCE) layer rather than a Softmax classifier.

2) Details: The attribute network shares some of its layers with the classification network. The convolutional and the fully connected layers of the attribute network are structurally similar with those of the classification network. However, the loss function is $Sigmoid$ rather than $Softmax$ of the classification network because the attribute learning is viewed as a multilabel classification (MLC) problem and is addressed based on binary relevance (BR). Given the last layer features $\mathbf{S}^L$

$$\mathbf{S}^L = \left\{\mathbf{S}_m^L | m = 1, 2, \ldots, M\right\}, \quad \mathbf{S}_m^L = \left[t_1^m, t_2^m, \ldots, t_D^m\right]^{\mathrm{T}} \tag{7}$$

the attribute prediction for the mth image is formulated as

$$\mathbf{A} = \mathrm{Sigmoid}\left(\mathbf{S}^L\right) = \{\mathbf{a}_m, | m = 1, 2, \ldots, M\} \tag{8}$$

where $\mathbf{a}_m = [a_1^m, a_2^m, \ldots, a_D^m]^{\mathrm{T}}$ represents the predicted probabilities for all the attributes. Computation of elements in $\mathbf{a}_m$ is formulated as

$$a_d^m = \mathrm{Sigmoid}\left(\mathbf{S}_m^L\right) = \frac{1}{1 + e^{-t_d^m}}. \tag{9}$$

During the training phase, the attribute network optimizes the loss function

$$\begin{aligned} L_{\mathrm{atr}} &= \mathrm{BinaryCrossEntropy}(\mathbf{A}) \\ &= \sum_m -\frac{1}{D}\sum_d \left(h_d^m \log\left(t_d^m\right) + \left(1 - h_d^m\right)\log\left(1 - t_d^m\right)\right). \end{aligned} \tag{10}$$

In this loss function, h_d^m is the ground truth label of the dth attribute of the mth image.

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D. Relationship Branch

In order to explore the latent relationship between these two tasks, we propose the relationship learning between two different tasks. As aforementioned, shallow layer features can be easily shared, while deep layer high-level semantic features sharing would be rather difficult. In order to address this problem, we adopt the soft-parameter-sharing way (see Section III-E) for deep layer features. We, in detail, map the high-level semantic features of the two tasks to the same feature space and minimize the distance between the mapping features of the two tasks. Through this mapping relationship, a deep latent relationship between tasks can be reflected. Viewing MTL as inductive transfer, the attribute task provides the inductive bias for the classification task [18], using the deep hidden relationship obtained from relationship learning.

1) Overall Description: Since the relationship between the fully connected layers of different tasks is weak, a soft-parameter-sharing strategy is utilized. A regularization constraint is proposed on the distances between different model parameters. Through this constraint, the fully connected layers of different branches are expected to be as similar as possible.

As shown in the right part of Fig. 3, the purple stream denotes the relationship branch. This branch starts with two feature vectors produced by FC^c and FC^a, which respectively correspond to the classification branch and the attribute branch. Since high-level features of different branches are hard to share [18], the feature vectors FC^c and FC^a should first be mapped to the same feature space. Therefore, FC^3 and FC^4 are implemented to get the mapped features.

2) Details: Start with the deep classification features $\mathbf{I}^L$ and the deep attribute features $\mathbf{S}^L$, we first take two fully connected layers to transfer these features into the same feature space

$$\tilde{\Gamma} = \mathrm{Activate}\left(\mathbf{W}^{\Gamma}\Gamma + \mathbf{B}^{\Gamma}\right)|_{\Gamma=\mathbf{I}^L,\mathbf{S}^L}. \tag{11}$$

After this, we can get $\tilde{\mathbf{I}}^L$ and $\tilde{\mathbf{S}}^L$. Then the relationship learning constraint is formulated as

$$L_{\mathrm{rlsn}} = \left\|\tilde{\mathbf{I}}^L - \tilde{\mathbf{S}}^L\right\|^2. \tag{12}$$

Therefore, the relationship loss term L_{rlsn} would be a positive value.

E. Synthetic Weights Sharing Mechanism

As shown in Fig. 4, in the classification branch and the attribute branch, weights of the same depth convolutional layers are shared. It should be noted that fully connected layers between the two networks are not shared. Investigations have been made on the NN weights sharing between different tasks [28]–[31]. First and foremost, there is a commonly acknowledged point in these methods that deep layer features are harder to share than the shallow layer features [18]. Besides, these works have also indicated that whether the weights are sharable or not depends on task similarity [29]. Thus, we design a synthetic weights sharing mechanism, and the processes are shown as follows.

1) Hard Sharing Mechanism: Currently, MTL based on DNNs can be divided into two categories according to the sharing mechanism: soft parameter sharing and hard parameter sharing. Hard parameter sharing is the most common method in DNN, which makes several networks adopt the same set of parameters directly. Such settings allow these shared parameters to have the potential to express multiple tasks, reduce overfitting risks, and improve generalization performance. In soft parameter sharing, different tasks have different groups of parameters, and the similarity of parameters is ensured by applying regularization constraints on the distance of model parameters.

However, in most cases, deep semantic features between different tasks are difficult to share directly, as described in the previous analysis. To make it clearer, we have investigated an example of deep semantic features sharing, shown in the following. This framework is named as "attribute hard-shared," visualized in Fig. 5(a). As shown in this figure, both convolutional features and deep fully connected features are hard-shared for different tasks. This way of directly sharing parameters of deep layers has neglected the fact that different task deep features vary a lot.

To address this deep-unsharable problem, "attribute separated" is applied in this article, where no hard sharing mechanism is adopted for deep semantic features. The "attribute separated" sharing mechanism is also named as $\mathrm{ACCNN}^{\mathrm{A}}$. This kind of sharing mechanism is shown in Fig. 5(b). It can be seen in this figure that the fully connected features are separated.

2) Soft Sharing Mechanism: Although "attribute separated" seems to address the deep-unsharable problem well, there is another problem. Deep semantic features are unsharable in a hard-parameter-sharing way, but may be sharable in other ways where these deep semantic features are separated. These two methods are investigated in the following.

The first one is named as "relationship adding-based." Its convolutional layers are hard-shared and its fully connected layers are separated. This framework is visualized in Fig. 5(c), where the attribute deep features are added to the classification deep features in an elementwise way. In fact, this framework fails to make the auxiliary task improve the major task well. This phenomenon will be proved in our experiments.

Then, the second method is the proposed "relationship soft-shared," namely, the ACCNN. The proposed novel parameter sharing mechanism is visualized in Fig. 5(d). Considering the weak relationship between deep semantic features of different tasks, the framework with a soft-parameter-sharing mechanism on fully connected layers is proposed. Detailedly, as shown in III-D2, this soft-parameter-sharing mechanism is implemented by a regularization constraint on different branch parameter distances.

IV. Experiments

In this section, a pipeline is first proposed to build attribute cooperated classification (ACC) data sets. Then three attribute-classification data sets built by this pipeline are illustrated. Finally, different kinds of experiments are conveyed to evaluate the validity of our model.

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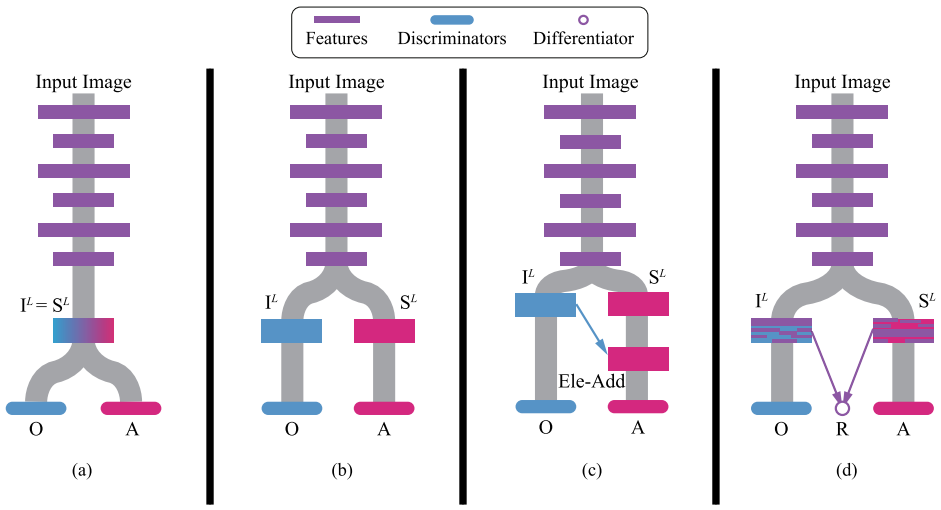


Fig. 5. Illustration of the investigated frameworks. Three kinds of parameter sharing mechanisms are listed and the last one is the proposed. The data spreading is shown in a broad gray line. **C**, **A**, and **R** represent the classification task, the attribute task, and the relationship between these tasks, respectively. $\mathbf{I}^L$ denotes the last layer of the classification branch. $\mathbf{S}^L$ refers to the last layer of the attribute branch. Colors of rectangles: different task features. Blue: main task features. Red: auxiliary task features. Purple: shared features. (a) Attribute hard-shared. (b) Attribute separated (ACCNNA). (c) Relationship adding-based. (d) Relationship soft-shared (ACCNN).

A. Pipeline of Building Data Sets

In order to avoid consuming quantities of time and efforts, a pipeline is designed to build attribute-classification data sets. Generally speaking, attribute is an abundant description of an image, containing plenty of different kind information in a short word vector. Our labeling method of an attribute is based on existing data sets with both classification and caption data sets. Detailed illustrations about the attribute and our labeling method are shown in the following.

1) Caption Data Sets: Caption labels of an image are several unstructured sentences describing various kinds of information. The criteria of building caption data sets are listed as follows [58].

1) All the important parts must be mentioned.
2) When multiple objects are in an image, never start the sentences with "There is."
3) In the absence of contrast, never use vague words like large, tall, and many.
4) Never use direction nouns like "west."
5) No less than six words should be contained in a sentence.

2) Attribute Labeling: To get attribute labels, function words are filtered out from caption labels. Then the most frequent words are saved with a ratio. Details are shown in the following.

1) Searching for appropriate caption data sets which have classification labels. In this step, three data sets are found: RSICD [58], UCM-caption [59], and Sydney-caption [59].
2) There are five sentences describing an image in a caption data set. Therefore, delete the redundant words and keep a nonrepeating "word_vector" for each image.
3) For each of the caption data sets, count word quantities in the "word_vectors" of all images. Save the results to a list.
4) From the list, exclude words that are subjective or belong to function words (pronouns, interjections, onomatopoeia, prepositions, conjunctions, and auxiliaries).
5) Rank the new list by quantities and save the most frequent words with a ratio to a nonrepeating "attribute_list."
6) For each image, attach a binary vector with the same length as "attribute_list." Elements of this vector represent the existence of different attributes.

TABLE I
RELATED INFORMATION ABOUT AC-AID

		Value	
NO. of Images		10000	
Size of Images		600×600	
NO. of Classes		30	
NO. of Attributes		83	
	Min	Average	Max
NO. of Images per Class	220	331	420
NO. of Images per Attribute	179	767	5899

B. Data Sets

In order to evaluate the proposed method, three attribute data sets are built based on the existing classification data sets and caption data sets. In this section, overall illustrations on the three data sets are shown in three tables. Detailed attribute label types and their counts are provided on the website mentioned earlier.

1) Attribute Cooperated-AID (AC-AID): Based on the classification data set AID [60] and the caption data set RSICD [58], we build a novel data set AC-AID, which is an MTL data set of classification and attribute learning.

As an MTL data set, information of images, attribute labels, and scene category labels are illustrated in Table I. As shown in this table, this data set contains 10 000 images of

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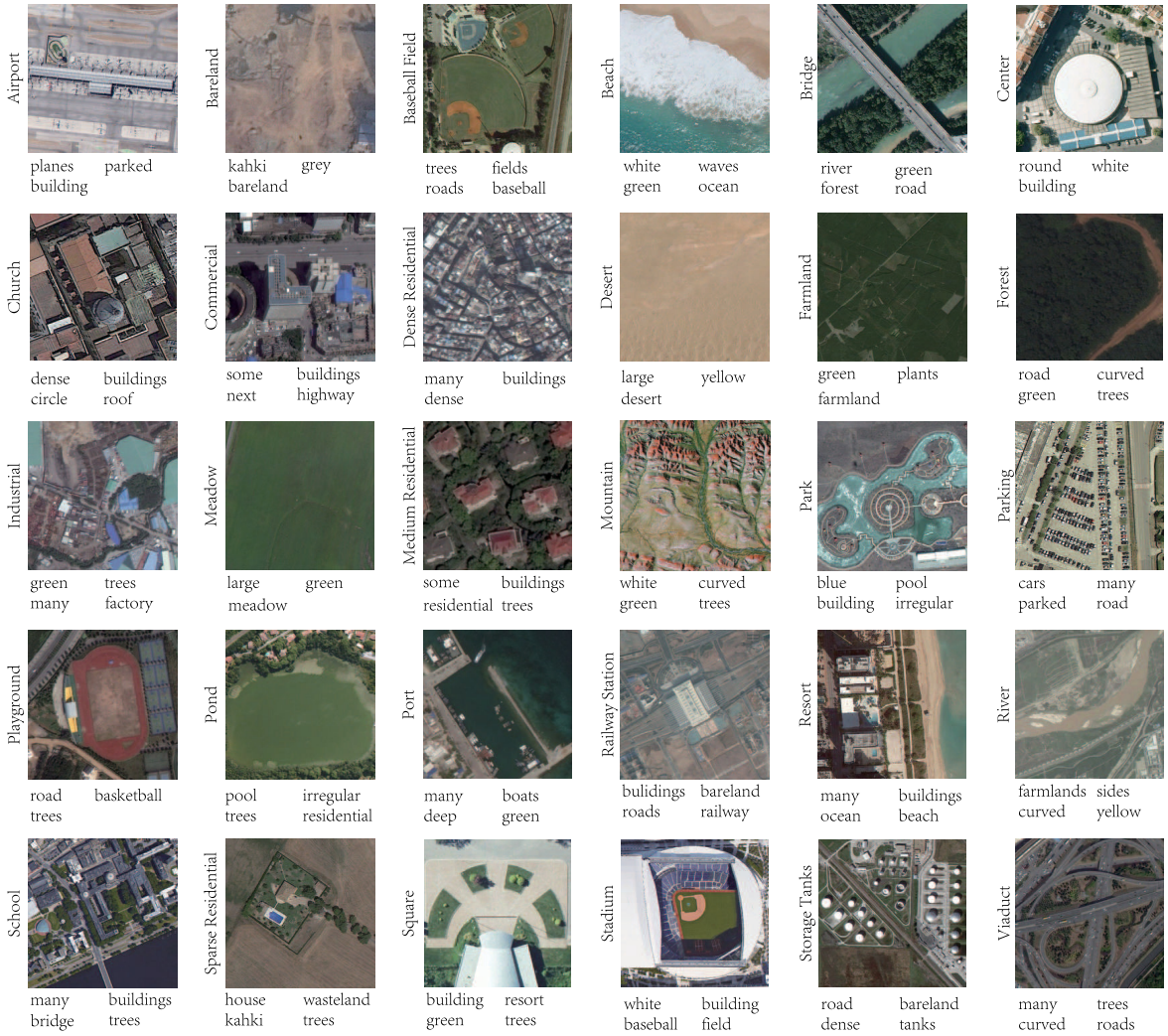


Fig. 6. Samples of the MTL data set AC-AID. Classification labels are shown on the left side of the images. Several elements of attribute label vectors are shown below the images.

size 600×600. These images are divided into 30 classes with 83 attributes. For each class, there are averagely 331 samples. For each attribute, there are averagely 767 images containing this attribute.

Samples of this data set are shown in Fig. 6. Classification labels are shown on the left side of the images. Several elements of attribute label vectors are shown below the images.

2) Attribute Cooperated-UCM (AC-UCM): Based on the classification data set UCM [8] and the caption data set UCM-caption [59], we build a novel data set (AC-UCM), which is an MTL data set of classification and attribute learning.

As an MTL data set, information of images, attribute labels, and scene category labels are illustrated in Table II. As shown in this table, this data set contains 2100 images of size 256×256. These images are divided into 21 classes with 56 attributes. For each class, there are 100 images. For each attribute, there are averagely 228 images containing this attribute. Samples of this data set are shown in Fig. 7.

3) Attribute Cooperated-Sydney: Based on the classification data set Sydney [61] and the caption data set Sydney-caption [59], we build a novel data set *Attribute Cooperated-Sydney* (AC-Sydney), which is an MTL data set of classification and attribute learning.

As an MTL data set, information of images, attribute labels, and scene category labels are illustrated in Table III. As shown in this table, this data set contains 2100 images of size 500×500. These images are divided into seven classes with 30 attributes. For each class, there are averagely 88 images. For each attribute, there are averagely 167 images containing this attribute. Samples of this data set are shown in Fig. 8.

TABLE II
RELATED INFORMATION ABOUT AC-UCM

	Value		
NO. of Images	2100		
Size of Images	256×256		
NO. of Classes	21		
NO. of Attributes	56		
	Min	Average	Max
NO. of Images per Class	100	100	100
NO. of Images per Attribute	87	228	1205

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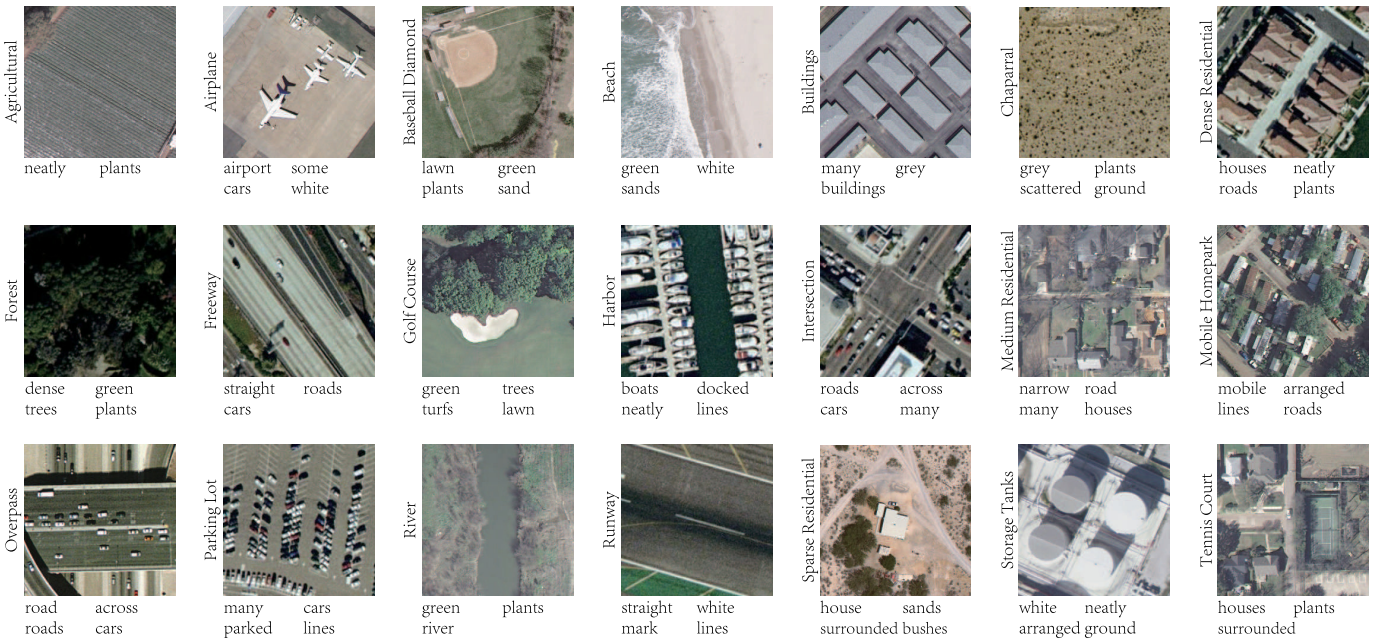


Fig. 7. Samples of the MTL data set AC-UCM. Classification labels are shown on the left side of the images. Several elements of attribute label vectors are shown below the images.



Fig. 8. Samples of the MTL data set AC-Sydney. Classification labels are shown on the left side of the images. Several elements of attribute label vectors are shown below the images.

TABLE III
RELATED INFORMATION ABOUT AC-SYDNEY

		Value	
NO. of Images		613	
Size of Images		500×500	
NO. of Classes		7	
NO. of Attributes		30	
	Min	Average	Max
NO. of Images per Class	22	88	242
NO. of Images per Attribute	47	167	442

C. Experimental Settings

Before the experiments, experimental settings are illustrated from these aspects: image preprocessing, data augmentation, parameter settings, and evaluation metrics.

1) Image Preprocessing: While implementing the experiments, a problem of computing resource shortage always occurs on the AID data set and the Sydney data set. After several analyses and experiments, the reason is found to be the oversize images which cause large requirements of computing resources. Therefore, before any processing, images from AID and Sydney are resized to 224×224, which is the size of UCM images.

2) Data Augmentation: It is well known that the overfitting problem usually happens in DNN training. In order to cope with this problem, various data augmentation methods have been proposed in the paper of ImageNet [41], such as random cropping, flipping, and PCA Jittering. Out of these methods, we select the random cropping method and the random horizontal flipping method for data enhancement.

a) Random cropping: As shown in Fig. 9(a), the random cropping method includes two steps. First, this method makes padding around an image. Then, it generates new samples with a fixed size at random center positions. In our experiments, the padding size is set to $1/8$ of the input image size (width equals height), and the cropping size is set the same as the input image size.

b) Random horizontal flipping: As shown in Fig. 9(b), the random flipping method first randomly selected half of the input images. Then, the horizontal flipping operation is implemented on these images. The flipped images and the other images are taken as the augmented new data set.

3) Parameter Setting: Besides image preprocessing and data augmentation, parameter setting is another important part to improve model performance.

a) Parameter illustration: As shown in Fig. 3, the proposed method contains several convolutional layers and three branches: classification, attribute, and relationship. The three branches correspond to different tasks and need different

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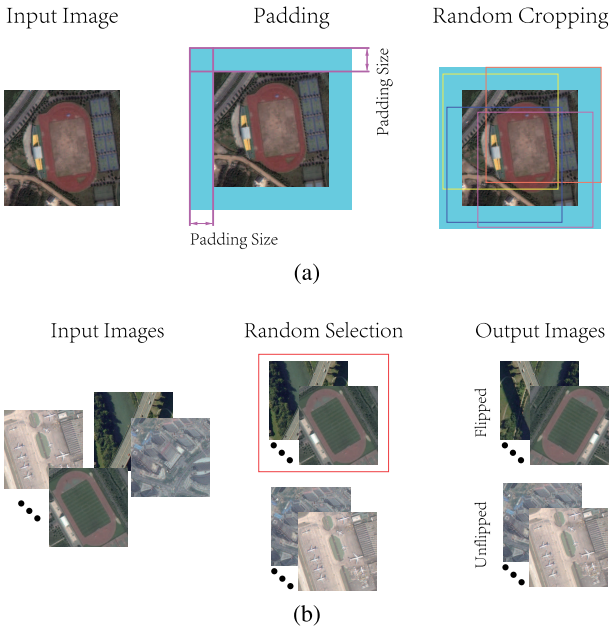


Fig. 9. Illustration of the adopted data augmentation methods. (a) Random cropping: after padding around the image, crop with fixed size and random position. (b) Random horizontal flipping: randomly choose half of the images to flip horizontally.

TABLE IV
PARAMETER SETTINGS OF THE PROPOSED MODEL

Item	Value
bLR	0.01
bWD	0.01
λ_a	15
$\mathrm{LR_a}$	0.005
$\mathrm{WD_a}$	0.1
λ_r	0.001
$\mathrm{LR_r}$	0.005
$\mathrm{WD_r}$	0.1

parameter settings. Learning rates of different parts are represented as: bLR for the convolutional layers and the classification branch; $\mathrm{LR_a}$ for the attribute branch; and $\mathrm{LR_r}$ for the relationship branch. Weight decays of different parts are represented as: bWD for the convolutional layers and the classification branch; $\mathrm{WD_a}$ for the attribute branch; and $\mathrm{WD_r}$ for the relationship branch. Moreover, as shown in formula 1, λ_a and λ_r respectively represent the regularization parameters of the attribute branch and the relationship branch.

b) Detailed setting: In the training process of the proposed model, the parameters are listed in Table IV. All the parameters are validated on the middle-scale AC-UCM and kept fixed on AC-AID and AC-Sydney. In detail, parameter selection experiments are individually implemented on $\mathrm{LR_a}$, $\mathrm{LR_r}$, $\mathrm{WD_a}$, and $\mathrm{WD_r}$. Then λ_a is roughly determined for the selection of λ_r. After this, parameter selection experiments are implemented in the order of first λ_r then λ_a. The experimental results of λ_r and λ_a are shown in Table V. Finally, the momentum of the whole network is set to 0.9.

4) Evaluation Metrics: The proposed model consists of two tasks. Therefore, we adopt two groups of evaluation methods to verify the validation of our method.

TABLE V
PARAMETERS SELECTING ON AC-UCM DATA SET (TWOFOLDS)

λ_r (λ_a=15)	10^0	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
Err. Cls /%	—	—	4.29	**4.05**	4.76	5.29
Err. Atr /%	—	—	3.80	3.58	3.48	**3.42**
λ_a (λ_r=0.001)	1	10	15	20	30	40
Err. Cls /%	4.05	4.05	**4.05**	4.05	6.66	8.33
Err. Atr /%	8.30	3.72	**3.58**	3.65	4.18	4.74

TABLE VI
BINARY CONFUSION MATRIX

		Ground Truth	
		Positive	Negative
Prediction	Positive (**P**)	True Positive (**TP**)	False Positive (**FP**)
	Negative (**N**)	False Negative (**FN**)	True Negative (**FN**)

TABLE VII
ABLATION EXPERIMENTS ON AC-UCM DATA SET (TWOFOLDS)

	Attribute	Relationship	Err. Cls /%	Err. Att /%
VGG			7.14	
$\mathrm{ACCNN^A}$	✓		5.00	3.59
ACCNN	✓	✓	**4.05**	**3.58**

a) Classification branch: For the classification branch, we adopt error rate (Err.) as the evaluation metric. Given a confusion matrix in Table VI that illustrates the conception of TP, FP, FN, and TN, the definition of Err. can be formulated as

$$\mathrm{Err.} = \frac{\mathrm{FP} + \mathrm{FN}}{P + N}. \tag{13}$$

b) Attribute branch: For the attribute branch, we also adopt Err. as the evaluation metric. Moreover, the Err. of different attribute items are visualized in histograms, since there are too many attributes to show.

D. Contrast Methods

To evaluate the efficiency of the proposed method, contrast experiments will be implemented. According to whether VGG-Net is taken as the baseline, contrast methods are divided into two groups: VGG-Net-based methods and state-of-the-art methods. Details are shown as follows.

1) VGG-Net Based:

a) VGG-Net: The VGG-Net [42] has been proposed for the classification task of natural images. In the following experiments, the VGG-Net is treated as the baseline of the proposed method.

b) DCA-Fusion: DCA-Fusion has been proposed by Chaib *et al.* [52]. This method proposes to integrate the CNN feature and several selected fully connected layers with the DCA method.

2) State-of-the-Art:

a) Compressive sensing (CS)-Fusion: The CS-Fusion method has been proposed by Mekhalfi *et al.* [62]. This method takes CS to integrate different kinds of image features.

b) CNN-AE: In this method, Othman *et al.* [46] have tried an approach of taking advantage of different kinds of

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TABLE VIII
WEIGHTS SHARING MECHANISM EXPERIMENTS ON AC-UCM DATA SET (TWOFOLDS)

	Err. Cls /%	Err. Att /%
Attribute Hard-Shared	7.86	4.53
Attribute Separated (ACCNN^{A})	**5.00**	**3.59**
Relationship Adding-Based	10.00	4.53
Relationship Soft-Shared (ACCNN)	**4.05**	**3.58**

NNs. First, CNN is utilized to get the supervised feature representation. Then, AE is used to implement unsupervised feature representation on the CNN feature.

c) CNN-NN: The same as CNN-AE, CNN-NN is also proposed by Othman *et al.* [46]. This method replaces the sparse AE in CNN-AE with a latent NN.

d) Gradient boosting random convolutional network (GBRCN): Considering the development of all kinds of NNs, Zhang *et al.* [63] have proposed an effective combination method of different NNs, namely, the GBRCN.

After all these methods have been introduced, it is obvious that the previous methods mainly focus on feature fusion [46], [52], [62] or framework fusion [63] in the field of RSI classification. In the following experiments, contrast experiment results are from the work of Yuan *et al.* [64].

E. Results and Analysis

In this section, different methods illustrated in Section IV-D are implemented on different data sets, which have been introduced in Section IV-B. After this, evaluation metrics of different tasks, which have been formulated in Section IV-C, are calculated. The final results are shown in the following and the analysis have been made.

1) Ablation Experiments: To illustrate the effectiveness of each part of our framework, ablation experiments are implemented with the results shown in Table VII. It can be seen that the attribute branch reduces the classification Err. by 2.14%. Moreover, together with the relationship branch, the classification Err. is reduced by 3.09%. This result shows that both parts of the proposed framework contribute to the classification task.

2) Weights Sharing Mechanism Experiments: During investigation of the proposed framework, we have tried different kinds of weights sharing mechanisms, which are shown in Fig. 5. These different mechanisms are implemented on the AC-UCM data set, and the results are shown in Table VIII.

First, different attribute learning mechanisms are experimented. Compared with "attribute hard-shared," "attribute separated" used in our method reduces the Err. by 2.86%. This result shows that directly using hard parameter sharing for high-level features is harmful to the original task. The attribute branch and the classification branch should be separated to achieve efficiency.

Second, based on the aforementioned "attribute separated" mechanism, different relationship learning mechanisms are experimented. Compared with "relationship adding-based," "relationship soft-shared" used in our method reduces the Err. by 5.95%. This result shows that directly using adding operation for high-level features is also harmful to the original task.

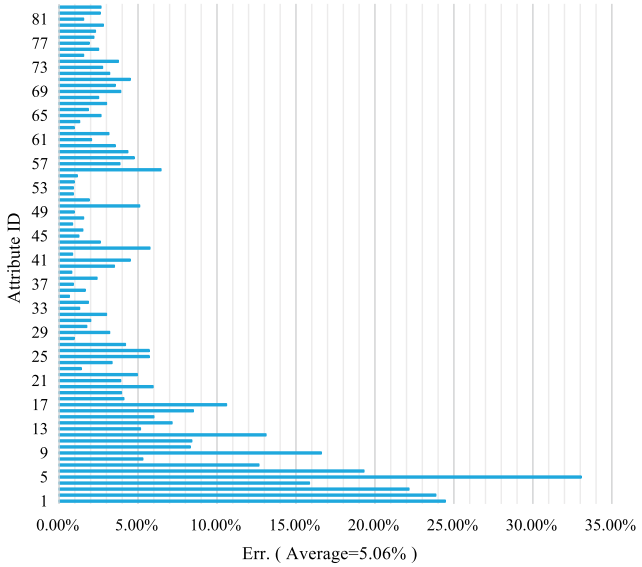


Fig. 10. Attribute error rate of ACCNN on the AC-AID data set. The "Attribute ID" denotes different attribute items.

TABLE IX
CLASSIFICATION ERROR RATES OF DIFFERENT METHODS ON THE THREE DATA SETS (TWOFOLDS)

	AC-AID Err. /%	AC-UCM Err. /%	AC-Sydney Err. /%
CS-Fusion	9.94	9.35	10.45
GBRCN	8.60	8.22	7.81
CNN-AE	7.97	7.40	7.17
CNN-NN	7.03	6.51	6.74
VGG-Net (base)	8.70	7.14	7.19
DCA-Fusion	**5.67**	6.16	5.99
ACCNN	5.87	**4.05**	**4.13**

3) Contrast Experiments:

a) AC-AID: The experimental results on AC-AID consist of the attribute prediction results and the classification results. The attribute prediction results are shown in Fig. 10, where the "attribute ID" denotes different attribute items. As shown in this figure, the Err.s of these items range from 0.65% to 33.05%. The average Err. of these items is 5.06%. This result shows that the attribute learning branch can work efficiently.

The classification results are shown in Table IX. Compared with the baseline VGG-NET with an Err. of 8.70%, the proposed ACCNN reduces the Err. by 2.83%–5.87%. This performance of ACCNN is the second best on AC-AID and is approximate to the best method, DCA-Fusion, which has an Err. of 5.67%. Therefore, the proposed method shows a competitive performance on AC-AID.

b) AC-UCM: Similar to experiments on AC-AID, the experimental results on AC-UCM consist of two parts described as follows. The attribute prediction results are shown in Fig. 11. As shown in this figure, the Err.s of these items range from 0.00% to 13.81%. The average Err. of these items is 3.58%. This result shows that the attribute learning branch can work efficiently.

The classification results on AC-UCM are illustrated in Table IX. Compared with the baseline VGG-NET with

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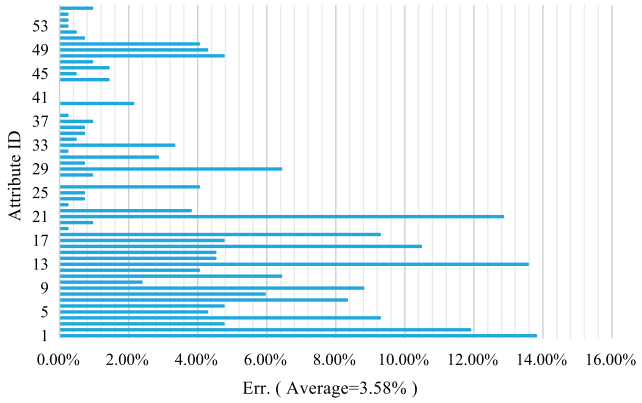


Fig. 11. Attribute error rate of ACCNN on the AC-UCM data set. The "Attribute ID" denotes different attribute items.

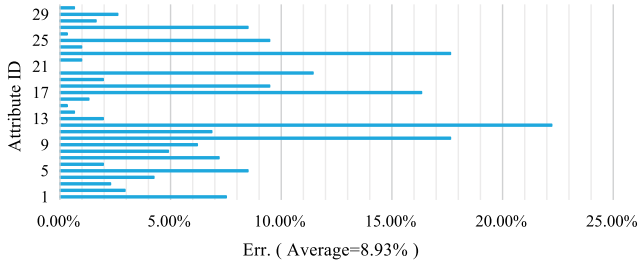


Fig. 12. Attribute error rate of ACCNN on the AC-Sydney data set. The "Attribute ID" denotes different attribute items.

an Err. of 7.14%, the proposed ACCNN reduces the Err. by 3.09%–4.05%. This performance of ACCNN is the best on AC-UCM. Moreover, this Err. is 2.11% lower than the Err. of the second-best method, DCA-Fusion, which gets an Err. of 6.16%. Therefore, the proposed method reaches a state-of-the-art performance on AC-AID.

c) AC-Sydney: The experimental results on AC-Sydney are illustrated in the following. First, the attribute prediction results are shown in Fig. 12. As shown in this figure, the Err.s of these items range from 0.00% to 22.22%. The average Err. of these items is 8.93%. This result shows that the attribute learning branch can work efficiently.

As it can be seen in Table IX, the classification result of ACCNN is the best on AC-Sydney. Compared with the baseline VGG-NET with an Err. of 7.19%, the proposed ACCNN reduces the Err. by 3.06%–4.13%. Moreover, this Err. is 1.86% lower than the Err. of the second-best method, DCA-Fusion, which gets an Err. of 5.99%. Thus, the proposed method reaches a state-of-the-art performance on AC-AID.

It can be seen that the proposed method outperforms most of the contrast methods. There are two reasons: the introduction of the attribute branch and the designing of weights sharing mechanism. First, information of both attribute and classification are considered in the proposed method, while only classification information is considered in the comparison methods. In the ablation experiments (see Table VII), it is proved that the attribute information is helpful to the proposed method. Second, a weights sharing mechanism is designed in the proposed method, since the attribute information is so different from the classification information. In the ablation experiments (see Table VII), it is proved that the proposed method can be further improved by the weights sharing mechanism.

V. Conclusion

In this article, we propose the ACCNN framework for remote sensing scene image classification. This framework is expected to address the problem of distinguishing visually fine-grained categories. This is because RSIs are captured from a top-down perspective, which would mix up some fine-grained categories. The proposed framework deals with this problem by taking attribute learning as an auxiliary task and adding a relationship constraint between the classification and the auxiliary task. The proposed ACCNN reduces the Err.s of the backbone VGG-Net by 2.83%, 3.09%, and 3.06% on AC-AID, AC-UCM, and AC-Sydney, respectively. Moreover, ACCNN reaches state-of-the-art performance on AC-UCM and AC-Sydney, and shows competitive performance with the state-of-the-art method on AC-AID. In addition, we build three MTL data sets AC-AID, AC-UCM, and AC-Sydney. They are respectively based on classical classification data sets AID, UCM, and Sydney.

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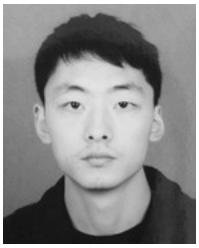
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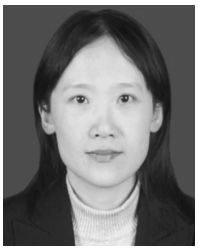
Yuanlin Zhang received the B.S. degree from the School of Instrumentation and Optoelectronic Engineering, Beihang University, Beijing, China, in 2015. He is pursuing the Ph.D. degree with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China.

He is also with the University of Chinese Academy of Sciences, Beijing. His research interests include deep learning and object detection on high-resolution remote sensing images.



Xiangtao Zheng (Member, IEEE) received the M.Sc. and Ph.D. degrees in signal and information processing from the University of Chinese Academy of Sciences, Beijing, China, in 2014 and 2017, respectively.

He is an Associate Professor with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, Shaanxi, China. His main research interests include computer vision and pattern recognition.



Yuan Yuan (Senior Member, IEEE) is a Full Professor with the School of Computer Science and the Center for OPTical IMagery Analysis and Learning (OPTIMAL), Northwestern Polytechnical University, Xi'an, China. She has authored or coauthored more than 100 articles, including over 70 in reputable journals, such as the IEEE TRANSACTIONS and *Pattern Recognition*, as well as conferences articles in the Conference on Computer Vision and Pattern Recognition, the British Machine Vision Conference, the International Conference on Image Processing, the International Conference on Acoustics, Speech, and Signal Processing, and so on. Her research interests include visual information processing and image/video content analysis.



Xiaoqiang Lu (Senior Member, IEEE) is a Full Professor with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China. His research interests include pattern recognition, machine learning, hyperspectral image analysis, cellular automata, and medical imaging.

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